Habitual physical exercise has beneficial effects on telomere length in postmenopausal women.

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OBJECTIVE: It has been reported that women benefit from the maintenance of telomere length by estrogen. Exercise may favorably influence telomere length, although results are inconsistent regarding the duration and type of exercise and the cell type used to measure telomere length. The purpose of this study was to investigate the relationship between habitual physical exercise and telomere length in peripheral blood mononuclear cells (PBMCs) in postmenopausal women. Postmenopausal women were chosen as study participants because they are typically estrogen deficient. METHODS: This experimental-control, cross-sectional study included 44 healthy, nondiabetic, nonsmoking, postmenopausal women. Habitual exercisers and sedentary participants were matched for age and body mass index. Body weight, height, blood pressure, and waist and hip circumference were measured. Mitochondrial DNA copy number and telomere length in PBMCs were determined, and biochemical tests were performed. Habitual physical exercise was defined as combined aerobic and resistance exercise performed for at least 60 minutes per session more than three times a week for more than 12 months. RESULTS: The mean age of all participants was 58.11 ± 6.84 years, and participants in the habitual exercise group had been exercising more than three times per week for an average of 19.23 ± 5.15 months. Serum triglyceride levels ($P = 0.01$), fasting insulin concentrations ($P < 0.01$), and homeostasis model assessment of insulin resistance ($P < 0.01$) were significantly lower and high-density lipoprotein cholesterol levels ($P < 0.01$), circulating adiponectin ($P < 0.01$), mitochondrial DNA copy number ($P < 0.01$), and telomere length ($P < 0.01$) were significantly higher in the habitual exercise group than in the sedentary group. In a stepwise multiple regression analysis, habitual exercise ($\beta = 0.522$, $P < 0.01$) and adiponectin levels ($\beta = 0.139$, $P = 0.03$) were the independent factors associated with the telomere length of PBMCs in postmenopausal women. CONCLUSIONS: Habitual physical exercise is associated with greater telomere length in postmenopausal women. This finding suggests that habitual physical exercise in postmenopausal women may reduce telomere attrition.